

SUDBURY, ON, Canada

WMO: 717300

Lat: 46.6237N

Lon: 80.7939W

Elev: 348

StdP: 97.21

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-24.8	-33.1	0.2	-27.7	-30.0	0.2	-24.2	11.4	-8.3	10.3	-8.0	3.1	330	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.0	20.1	27.3	19.0	25.7	18.1	21.6	26.6	20.5	24.9	19.5	23.6	5.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.3	23.4	19.0	14.4	22.7	18.1	13.6	21.7	64.5	26.7	60.7	24.8	57.4	23.6	25.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.9	8.0	DB	-31.6	31.9	3.3	1.9	-34.0	33.3	-35.9	34.4	-37.7	35.4	-40.1	36.8	(4)
(5)			WB	-31.8	23.6	3.2	1.1	-34.1	24.4	-36.0	25.0	-37.8	25.6	-40.2	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.5	-12.5	-10.7	-4.6	3.2	11.4	16.8	19.1	18.0	13.8	6.7	-0.8	-7.7	(6)	
	DBStd	12.23	7.51	6.33	6.39	5.12	4.72	3.91	3.04	3.14	4.31	4.70	5.45	6.48	(7)	
	HDD10.0	2994	697	578	453	211	40	2	0	0	13	127	323	549	(8)	
	HDD18.3	5182	955	812	711	454	221	73	27	44	144	362	573	807	(9)	
	CDD10.0	982	0	0	1	7	82	207	283	249	129	23	1	0	(10)	
	CDD18.3	129	0	0	0	0	5	28	52	34	10	1	0	0	(11)	
	CDH23.3	990	0	0	0	1	60	238	395	232	63	2	0	0	(12)	
	CDH26.7	209	0	0	0	0	12	52	91	45	9	0	0	0	(13)	
(14) Wind	WSAvg	3.9	4.1	4.2	4.2	4.4	4.1	3.5	3.3	3.3	3.7	4.1	4.1	4.0	(14)	
(15) Precipitation	PrecAvg	917	64	53	65	68	76	89	79	90	107	76	82	71	(15)	
	PrecMax	1120	126	117	127	181	145	195	200	200	244	152	174	104	(16)	
	PrecMin	670	18	14	25	19	22	37	29	22	26	9	16	30	(17)	
	PrecStd	114	26	28	30	33	33	44	44	44	50	29	34	19	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	5.0	13.8	21.5	28.4	30.4	31.6	30.3	27.9	22.1	13.7	7.1	(19)	
		MCWB	4.5	3.1	8.8	12.5	18.0	20.4	21.7	21.2	20.3	15.9	10.4	6.1	(20)	
	2%	DB	2.0	2.3	9.2	17.2	24.8	28.2	29.1	27.8	25.1	18.7	10.9	4.0	(21)	
		MCWB	1.3	1.1	5.4	9.4	15.5	19.2	20.5	19.5	18.7	14.3	9.2	3.3	(22)	
	5%	DB	0.2	0.6	6.3	13.9	22.3	26.1	27.2	25.9	22.8	16.2	8.6	2.3	(23)	
		MCWB	-0.4	-0.4	3.3	7.7	14.1	17.9	19.1	18.5	17.3	13.0	7.3	1.7	(24)	
	10%	DB	-1.3	-1.1	3.9	11.2	19.7	24.1	25.5	24.1	20.7	13.7	6.7	0.9	(25)	
		MCWB	-2.0	-2.1	1.3	6.0	12.8	16.7	18.3	17.7	16.4	11.4	5.5	0.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	3.8	10.4	14.1	19.8	22.4	23.4	22.4	21.5	17.3	11.7	6.3	(27)	
		MCDB	4.9	4.6	11.6	17.8	25.5	29.1	29.1	27.3	26.1	19.5	13.1	6.7	(28)	
	2%	WB	1.5	1.4	6.1	10.9	17.5	20.7	21.7	21.0	19.8	15.2	9.5	3.5	(29)	
		MCDB	1.9	2.2	8.3	14.8	22.2	25.8	26.7	25.1	22.9	17.8	10.4	4.0	(30)	
	5%	WB	-0.3	-0.3	3.6	8.5	15.7	19.4	20.5	19.9	18.6	13.4	7.5	1.7	(31)	
		MCDB	0.2	0.5	6.0	13.0	19.8	24.0	25.0	23.8	21.5	15.7	8.5	2.2	(32)	
	10%	WB	-2.0	-2.0	1.6	6.3	13.8	18.1	19.5	18.9	17.3	11.6	5.5	0.3	(33)	
		MCDB	-1.3	-1.1	3.5	10.5	18.4	22.1	23.6	22.5	20.3	13.6	6.6	0.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	9.5	9.7	9.9	11.1	10.6	10.5	10.0	9.7	7.8	6.7	7.1	(35)	
		MCDBR	7.8	8.5	11.0	14.3	14.3	13.0	12.8	12.2	11.9	11.1	9.2	6.8	(36)	
	5% WB	MCWBR	7.8	7.8	7.8	8.6	7.4	6.1	5.9	5.5	6.3	7.2	7.9	6.6	(37)	
		MCWBR	7.8	8.2	10.0	13.1	12.4	10.9	10.9	10.1	10.2	10.0	8.9	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.265	0.282	0.302	0.343	0.362	0.387	0.389	0.380	0.357	0.325	0.298	0.264	(40)	
		taud	2.312	2.243	2.315	2.366	2.391	2.343	2.362	2.381	2.441	2.517	2.511	2.406	(41)	
	Ebn at Noon		834	889	925	913	903	878	871	865	857	839	799	801	(42)	
		Edh at Noon	74	99	108	114	115	122	118	111	96	76	63	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.27	2.28	3.71	4.58	5.36	5.71	5.74	4.82	3.72	2.22	1.22	0.95	(44)	
	RadStd		0.19	0.28	0.34	0.40	0.47	0.45	0.38	0.32	0.37	0.27	0.15	0.09	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page